

ALLIANCE GIRLS 2025 PREDICTION



POSSIBLE KCSE EXAM CHEMISTRY

233/3

(PRACTICAL)

PAPER 3

TIME: 2¼ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

INSTRUCTIONS TO CANDIDATES

- a) Write your name and ADMISSION numbers in the spaces provided above.
- b) Sign and write the date of examination in the spaces provided above.
- c) Answer **ALL** the questions in the spaces provided in the question paper.
- d) Mathematical tables and silent electronic calculators may be used.
- e) All working **MUST** be clearly shown where necessary.
- f) Candidates should check the question paper to ascertain that **all** pages are printed as indicated and that no questions are missing.

For Examiner's Use Only

Question	Maximum Score	Candidate's Score
1	20	
2	10	
3	10	
TOTAL SCORE	40	

1. (20 marks) You are provided with:

- Solution **P**, hydrochloric acid
- Solution **Q**, containing 8.8g per litre of sodium hydroxide.
- 0.6g of an impure carbonate, solid **B**

You are required to determine the:

- a) Concentration of solution **P** in moles per litre.
- b) Percentage purity of the carbonate, solid **B**.

Procedure I

Fill the burette with sodium hydroxide, solution **Q**. Pipette 25.0cm³ of hydrochloric acid, solution **P** into a conical flask. Add 2 – 3 drops of methyl orange indicator and titrate. (The colour of the indicator changes from pink to yellow). Record your results in table 1 below. Repeat the titration two more times and complete the table.

Table I

	I	II	III
Final burette reading			
Initial burette reading			
Volume of solution Q used (cm ³)			

(4 marks)

a) What is the average volume of solution **Q** used?

(1 mark)

b) Determine the:

- (i) Concentration of solution **Q** in moles per litre. (Na = 23.0; O = 16.0, H = 1.0) (1 mark)

(ii) Concentration of solution **P** in mole per litre

(2 marks)

Procedure II

Using a 100ml measuring cylinder, measure out 100cm³ of solution **P** into a 250cm³ beaker.

Add all of solid **B** into the beaker containing solution **P**. Swirl the mixture and allow the reaction to proceed for about 4 minutes.

Label the solution obtained here as solution **K**.

Fill the burette with sodium hydroxide, solution **Q**. Pipette 25.0cm³ of solution **K** into a conical flask. Add 2- 3 drops of methyl orange indicator and titrate. Record your results in table II below. Repeat the titration two more times and complete the table.

Table II

	I	II	III
Final burette reading			
Initial burette reading			
Volume of solution Q (cm ³)			

(4 marks)

c) What is the average volume of solution **Q** used?

(1 mark)

d) Calculate the :

i. Moles of hydrochloric acid in 25.0cm³ of solution **K**.

(1 mark)

ii. Moles of hydrochloric acid in 100cm³ of solution **K**. (1 mark)

iii. Moles of hydrochloric acid in 100cm³ of the original hydrochloric acid solution **P**.
(1 mark)

iv. Moles of hydrochloric acid that were used up in the reaction with solid **B**. (1 mark)

v. Moles of the carbonate that reacted with hydrochloric acid. (1 mark)

e) Given that the relative formula mass of the carbonate is 72, calculate the:

i. Mass of the carbonate that reacted. (1 mark)

ii. Percentage purity of the carbonate, solid **B**. (1 mark)

2. (10marks) You are provided with solid **M**. Carry out the tests below and record your observations and inferences in the spaces provided.

a) Place **all** of solid **M** in a boiling tube and add about 10cm³ distilled water and shake.

Observations	Inferences

(½mark)

(½ mark)

b) Divide the solution into five portions of about 2cm³ each.

i. To the first portion, add aqueous sodium hydroxide dropwise until in excess.

Observations	Inferences

(1mark)

(1mark)

ii. To the second portion, add aqueous ammonia dropwise until in excess.

Observations	Inferences

(1mark)

(1mark)

- iii. Dip a glass rod in the third portion and heat it on a Bunsen burner flame, identify the colour of the flame produced.

Observations	Inferences

(½mark)

(½mark)

- iv. To the fourth portion, add about 1 cm³ of lead (II) nitrate solution.

Observations	Inferences

(1mark)

(1mark)

- v. To the fifth portion, add about 1 cm³ of dilute sodium hydroxide followed by a small piece of aluminium foil. Warm the mixture gently and carefully. Test any gas produced using blue and red litmus papers.

Observations	Inferences

(1mark)

(1mark)

3. (10marks) You are provided with solid **W**. Carry out the tests below and record your observations and inferences in the spaces provided.

a) Place about one third of solid **W** on a **metallic** spatula and burn it using a Bunsen burner.

Observations	Inferences

(1mark)

(1mark)

b) Place the remaining solid **W** into a boiling tube. Add 10cm³ of distilled water and shake well. Use 2cm³ portions of the mixture for each of the following reactions.

(i) To the first portion, add 2 drops of acidified potassium manganate (VII) and shake well.

Observations	Inferences

(1mark) (1mark)

(ii) To the second portion, add three drops of bromine water

Observations	Inferences

(1mark) (1mark)

- (iii) To the third portion put universal indicator paper.

Observations	Inferences

(1mark)

(1mark)

- (iv) To the fourth portion add a little solid Y, sodium hydrogen carbonate.

Observations	Inferences

(½mark)

(½mark)

- (v) To the fifth portion add about 2cm³ of ethanol and warm the mixture.

Observations	Inferences

(½mark)

(½mark)